# Sanskar Jain

Backend Engineer – Distributed Systems & Streaming
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## Summary

Backend engineer with 2+ years building high-throughput, event-driven distributed systems at Verisk Analytics, a compliance-heavy insurance/claims data platform. Strong CS fundamentals in data structures, algorithms, and system design, with hands-on **Spring Boot** and **Kafka**-based microservice experience. Deep grounding in consistency, idempotency, retries/backoff, DLQ handling, and horizontal scaling, with direct ownership of production reliability and incident diagnosis. Proven technical communicator who authors design docs, drives architecture reviews, and mentors on production-readiness.

## Skills

**Languages:** Java, C#, Python
**Backend:** Spring Boot, ASP.NET Core, REST APIs, Microservices, CQRS, Event-Driven Architecture, API Versioning
**Distributed Systems:** Kafka, AWS SQS/SNS, Idempotency, Retry/Backoff, DLQ, Partitioning, Backpressure, Eventual Consistency, Horizontal Scaling
**Cloud & DevOps:** AWS (Lambda, EKS, ECS, S3, ElastiCache), Docker, Kubernetes, Terraform, CI/CD (Azure DevOps)
**Observability & Ops:** OpenTelemetry, Splunk, Production Incident Diagnosis, Postmortems, Swagger/OpenAPI
**Databases:** MongoDB, PostgreSQL, MySQL, Redis, Indexing, Query Optimization
**Security & Tools:** OAuth 2.0, JWT, Okta, Kong Gateway, xUnit/NUnit, Jira

## Experience

**Verisk Analytics** Jan 2024 – Present
**Software Engineer II, Hyderabad, India**

- Co-designed **RulesRunner**, a stateless microservice using **CQRS** and **SQS** that offloaded compliance checks from the client, cutting wait time from **3000ms to 500ms (83%)** while enabling horizontal scaling under variable load.
- Architected an **event-driven ingestion pipeline** with idempotent message processing, retry/backoff strategies, and **DLQ**-backed **AWS Lambda** consumers, guaranteeing zero message loss and consistent state under partial failures.
- Authored technical design documents defining service boundaries, API contracts, partitioning strategy, and **SNS**-based pub/sub patterns consumed by multiple downstream distributed services.
- Engineered scalable ingestion/export backend for **Xactimate** handling **30+ data types** via async processing and **S3** streaming, reducing transfer time **38%** and data issues **25%** in a compliance-driven insurance data domain.
- Built **Spring Boot**-compatible REST API design patterns and **ASP.NET Core** services handling **200K+ weekly requests**, with structured error handling, input validation, and **Swagger/OpenAPI** contracts.
- Established **OAuth 2.0** authentication on **AWS EKS** behind **Kong Gateway**, integrating **Okta JWT** tokens for secure cross-service authorization across the distributed platform.
- On-call owner for production services: triaged incidents, drove root-cause postmortems, and improved API response time **34%** via **Redis** caching, connection pooling, and hot-path query optimization.
- Optimized **database indexing/partitioning** on **MongoDB** and **PostgreSQL** and authored idempotent **migration scripts**, sustaining performance and zero-downtime deployments at scale.
- Slashed deployment time **50%** and build cycles **33%** by re-architecting **CI/CD** pipelines with staged gates, parallel jobs, and automated tests; instrumented **OpenTelemetry**/**Splunk** to cut incident MTTD.
- Cut technical debt **55%** and improved runtime performance **30%** by migrating legacy services to modern async, resilience-first patterns; led code reviews on architecture and edge-case handling.

## Projects

**Expense Tracker – Kafka-Based Microservices Platform** Aug 2023 – Dec 2023

- Built a **5+ microservice** system in **Spring Boot** using async **Kafka** topics/partitions for inter-service communication and a **Kong API Gateway** for routing, auth, and rate limiting.
- Containerized services with **Docker** and orchestrated on **Kubernetes**; load-tested **1000+ concurrent transactions** to validate throughput, retry behavior, and failure recovery.

## Certifications & Education

**AWS Certified Cloud Practitioner** – Amazon Web Services Oct 2024
**B.Tech in Computer Engineering** – Vishwakarma Institute of Technology, Pune 2024 — CGPA: 8.8